

1. A context free grammar can be recognized by
- a) **Push down automata**
 - b) 2 way linearly bounded automata
 - c) Both (a) and (b)
 - d) None of the mentioned

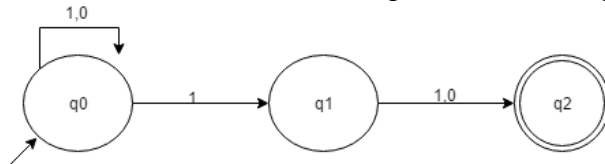
2. A Language for which no DFA exist is a_____
- a) Regular Language
 - b) **Non-Regular Language**
 - c) May be Regular
 - d) Cannot be said

3. Language of finite automata is.
- a) Type 0
 - b) Type 1
 - c) Type 2
 - d) **Type 3**

4. Moore Machine is an application of:
- a) Finite automata without input
 - b) **Finite automata with output**
 - c) Non Finite automata with output
 - d) None of the mentioned

5. Choose the correct option for the given statement:

Statement: The DFA shown represents all strings which have 1 at second last position.



- a) Correct
- b) **Incorrect, Incomplete DFA**
- c) Wrong proposition
- d) May be correct

6. Which one of the following is true?

A mealy machine_____

- a) produces a language
- b) produces a grammar
- c) can be converted to NFA
- d) **has less circuit delays**

7. The output alphabet can be represented as:

- a) δ
- b) Δ
- c) Σ
- d) $\square$

8. Statement 1: Mealy machine reacts faster to inputs.

Statement 2: Moore machine has more circuit delays.

Choose the correct option:

- a) **Statement 1 is true and Statement 2 is true**
- b) Statement 1 is true but Statement 2 is false
- c) Statement 1 is false and Statement 2 is true
- d) Statement 1 is false and Statement 2 is false

9. Assume the R is a relation on a set A, aRb is partially ordered such that a and b are

- a) reflexive b) transitive c) symmetric **d) reflexive and transitive**
10. Subset Construction method refers to:
a) Conversion of NFA to DFA b) DFA minimization
 c) Eliminating Null references d) ϵ -NFA to NFA
11. There are _____ tuples in finite state machines.
 a) 4 **b) 5** c) 6 d) 7
12. Transition function maps.
 a) $\Sigma * Q \rightarrow \Sigma$ b) $Q * Q \rightarrow \Sigma$ c) $\Sigma * \Sigma \rightarrow Q$ **d) $Q * \Sigma \rightarrow Q$**
33. The major difference between Mealy and Moore machine is about:
a) Output Variations b) Input Variations
 c) All of the mentioned d) None of the mentioned
14. The entity which generate Language is termed as:
 a) Automata b) Tokens **c) Grammar** d) Data
15. The production of form non-terminal $\rightarrow \epsilon$ is called:
 a) Sigma Production **b) Null Production**
 c) Unit Production d) All of the mentioned
16. Which of the following is an application of Finite Automaton?
 a) Compiler Design b) Grammar Parsers
 c) Text Search **d) All of the mentioned**
17. Which of the following statements is correct?
a) All Regular grammar are context free but not vice versa
 b) All context free grammar are regular grammar but not vice versa
 c) Regular grammar and context free grammar are the same entity
 d) None of the mentioned
18. The output alphabet can be represented as:
 a) δ **b) Δ** c) Σ d) None of the mentioned
19. Which among the following is not a part of the Context free grammar tuple?
a) End symbol b) Start symbol c) Variable d) Production
20. NFA, in its name has 'non-deterministic' because of :

- a) The result is undetermined b) The choice of path is non-deterministic
c) **The state to be transited next is non-deterministic** d) All of the mentioned

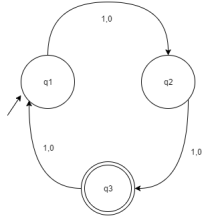
21. Which of the following is not an example of a finite state machine system?

- a) Control Mechanism of an elevator b) Combinational Locks
c) Traffic Lights **d) Digital Watches**

22. Which of the following is an application of Finite Automaton?

- a) Compiler Design b) Grammar Parsers c) Text Search **d) All of the mentioned**

23. Which of the following will the given DFA won't accept?



- a) ϵ b) 11010 c) 10001010 d) String of letter count 11

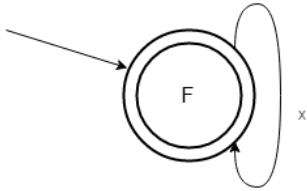
24. Which of the following statement is correct?

- a) All Regular grammar are context free but not vice versa
b) All context free grammar are regular grammar but not vice versa
c) Regular grammar and context free grammar are the same entity
d) None of the mentioned

25. Number of states required to accept a string ends with 10.

- a) 3 b) 2 c) 1 d) 4

26. What does the following figure most correctly represent?



- a) Final state with loop x
- b) Transitional state with loop x
- c) Initial state as well as final state with loop x**
- d) Insufficient Data

27. Which of the following is not an example of a finite state machine system?

- a) Control Mechanism of an elevator
b) Combinational Locks
c) Traffic Lights
d) Digital Watches

28.. What is wrong in the given definition of FA?

Def: ($\{q_0, q_1, q_2\}$, $\{0,1\}$, δ , q_3 , $\{q_3\}$)

- a) The definition does not satisfy 5 Tuple definition of NFA
- b) There are no transition definition
- c) Initial and Final states do not belong to the Graph**
- d) Initial and final states can't be same

29. Which of the following is an application of Finite Automaton?

- a) Compiler Design
- b) Grammar Parsers
- c) Text Search
- d) All of the mentioned**

30. Which of the following is not a part of 5-tuple finite automata?

- a) Input alphabet
- b) Transition function
- c) Initial State
- d) Output Alphabet**

31. An automaton that presents output based on previous state or current input:

- a) Acceptor
- b) Classifier
- c) Transducer**
- d) None of the mentioned.

32. NFA, in its name has 'non-deterministic' because of :

- a) The result is undetermined
- b) The choice of path is non-deterministic**
- c) The state to be transited next is non-deterministic
- d) All of the mentioned

33. Which of the following option is correct?

- a) NFA is slower to process and its representation uses more memory than DFA
- b) DFA is faster to process and its representation uses less memory than NFA
- c) NFA is slower to process and its representation uses less memory than DFA**
- d) DFA is slower to process and its representation uses less memory than NFA

34. What is the output for the given language?

Language: A set of strings over $\Sigma = \{a, b\}$ is taken as input and it prints 1 as an output "for every occurrence of a, b as its substring. (INPUT: abaaab)

- a) 0010001**

- b) 0101010
- c) 0111010
- d) 0010000

35. A regular language over an alphabet Σ is one that cannot be obtained from the basic languages using the operation

- a) Union
- b) Concatenation
- c) Kleene*
- d) All of the mentioned**